MARITIME COMPLIANCE TOOLS

USER GUIDE

FuelEU Maritime — Compliance Workbench

NavCIIFuel

Pooling, Scenario comparison, Pool agreement, Multi-year banking, Penalty calculator, Fuel comparison

Prepared for: Vessel Compliance & Reporting Teams

Document version: 1.0

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1 Introduction

1.1 Purpose of this guide

This guide explains how to use the FuelEU Maritime — Compliance Workbench, a browser-based tool for modelling compliance under the FuelEU Maritime regulation (EU 2023/1805). It walks through every tab, input field and result, using screenshots taken directly from the tool.

The guide is written for ship operators, technical superintendents, and compliance officers who need to pool vessel compliance balances, compare allocation strategies, draft a pooling agreement, bank or borrow balances across years, estimate penalty exposure, and compare fuel pathways against the regulatory GHG intensity target.

1.2 What the tool does

- Pools multiple vessels' adjusted compliance balances (tCO2eq) into a single fleet pool, allocating surplus from donor vessels to cover deficits.
- Generates four alternative allocation scenarios so you can compare different internal transfer strategies side by side.
- Drafts a formal Pooling Agreement document, pre-filled from your current pool results, ready for verifier and legal review.
- Tracks a pool's compliance balance across multiple years, applying the regulation's banking and borrowing rules automatically.
- Converts a residual compliance deficit into an estimated FuelEU penalty using the statutory VLSFO-equivalent formula.
- Compares the well-to-wake (WtW) GHG intensity of multiple fuel pathways against the compliance target for a chosen reporting year, with an interactive bar chart.

Before you start

- The tool runs entirely in your browser — no data is uploaded or sent anywhere.
- Have each vessel's adjusted compliance balance (tCO2eq), fuel consumption records, and reporting-year targets ready.
- Use a current version of Chrome, Edge, or Firefox for the best experience.

2 Interface Overview

The workbench is organised into a fixed masthead and six tabs, each covering one stage of the FuelEU compliance workflow: Pooling, Scenario comparison, Pool agreement, Multi-year banking, Penalty calculator, and Fuel comparison.

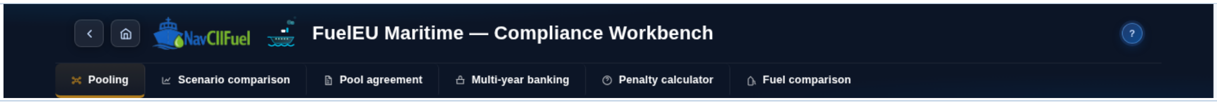


Figure 1 — Masthead with navigation, branding, tab bar, and the Page Help button.

The masthead is visible on every tab and provides the following controls:

Control	Function
Back / Home	The back arrow returns to the previous browser page; the home icon returns to the home page.
Tab bar	Switches between the six workbench tabs: Pooling, Scenario comparison, Pool agreement, Multi-year banking, Penalty calculator, Fuel comparison.
? (Page Help)	Opens the in-app help panel, with a quick-link table of contents and a description of every field on every tab.

3 Quick Start Workflow

Follow these steps for a typical single-pool, single-year compliance check:

Step	What to do
1	Open the workbench HTML file in your browser and confirm the Pooling tab loads with the fleet register table visible.
2	In the Pooling tab, add one row per vessel and enter its adjusted compliance balance (tCO2eq) — or click "Load sample fleet" to see the tool in action.
3	Review the KPI strip and the before/after allocation panels to see how surplus is distributed across the pool.
4	Switch to Scenario comparison and click "Generate all scenarios" to compare four different allocation strategies.
5	Switch to Pool agreement to generate a draft allocation record, fill in the metadata fields, and export to PDF.
6	Use Multi-year banking to carry balances across reporting years, and Penalty calculator to estimate any residual exposure.
7	Use Fuel comparison to check candidate fuel pathways against the target intensity for your chosen reporting year.

4 Pooling

The core workspace for combining several vessels into one compliance pool. Enter each vessel's name and its adjusted compliance balance (tCO2eq) — positive values are surplus, negative values are deficit.

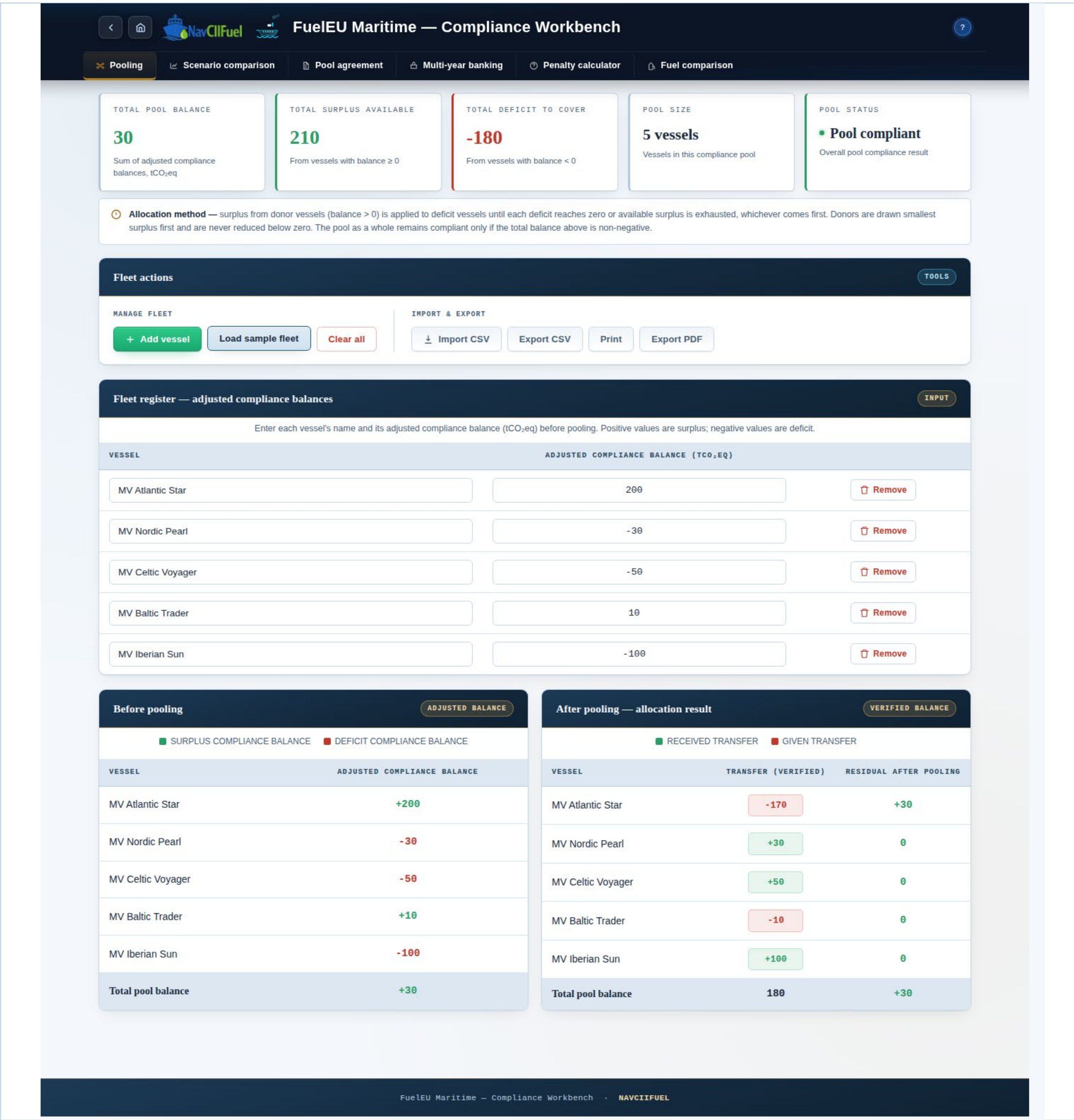


Figure 2 — Pooling tab with the sample fleet loaded, showing the KPI strip, fleet actions, and the fleet register table.

Field / Control	Description
Fleet register table	Enter each vessel's name and adjusted compliance balance. Add rows manually, load a sample fleet, import from CSV, or clear all rows.
KPI strip	Shows total pool balance, total surplus available, total deficit to cover, pool size (vessel count), and overall pool status at a glance.
Allocation method	Surplus from donor vessels (balance ≥ 0) is applied to deficit vessels until each reaches zero or the surplus runs out. Donors with the smallest surplus are drawn first, and no donor is reduced below zero. The pool is compliant overall only if the total balance is non-negative.
Initial vs. allocated panels	Two side-by-side cards show the fleet's balances before and after allocation, so you can see exactly how surplus moved between vessels.
Import & export tools	Import CSV, Export CSV, Print, and Export PDF let you bring in fleet data or take results out for filing and reporting.

5 Scenario Comparison

Click "Generate all scenarios" to automatically build four cards from your current fleet (entered on the Pooling tab), each applying a different internal transfer algorithm for distributing surplus to deficit vessels.

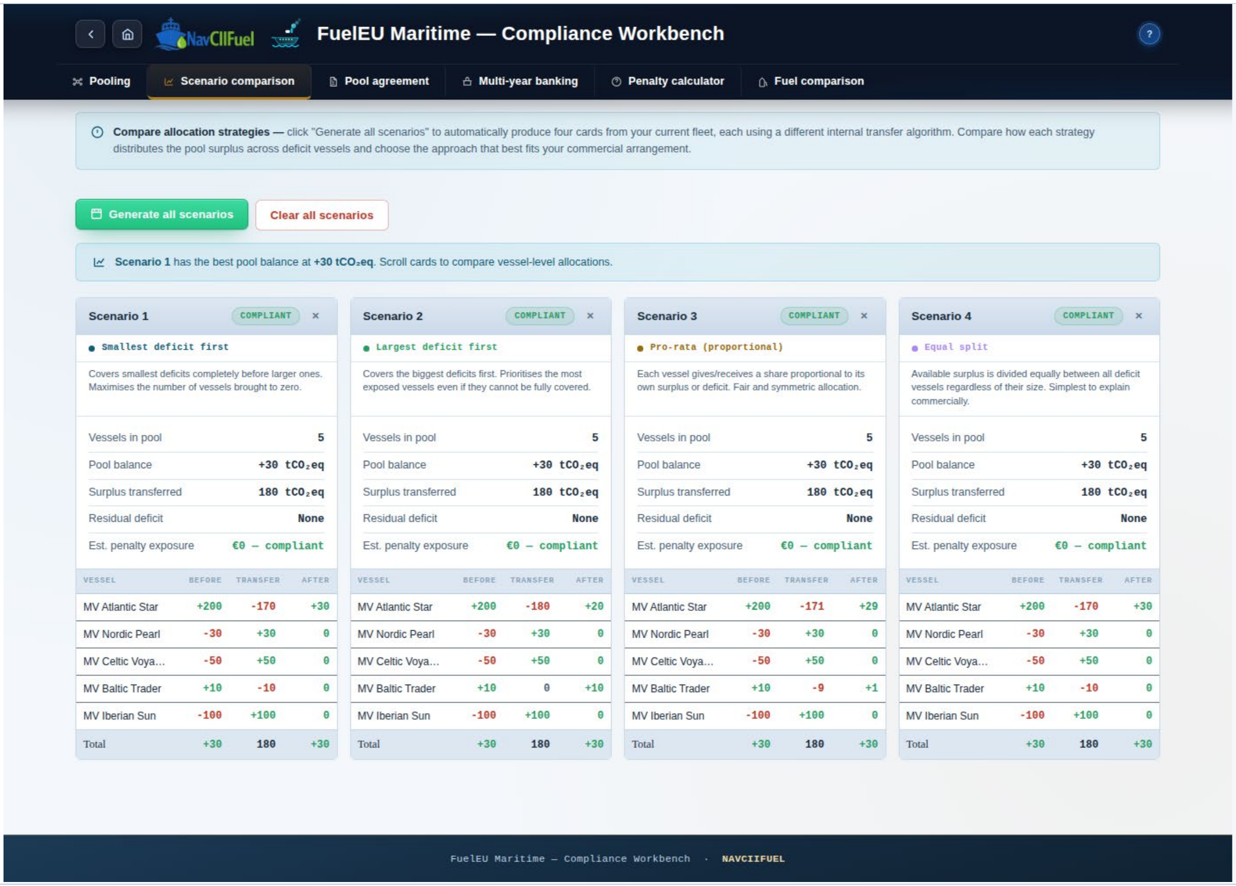
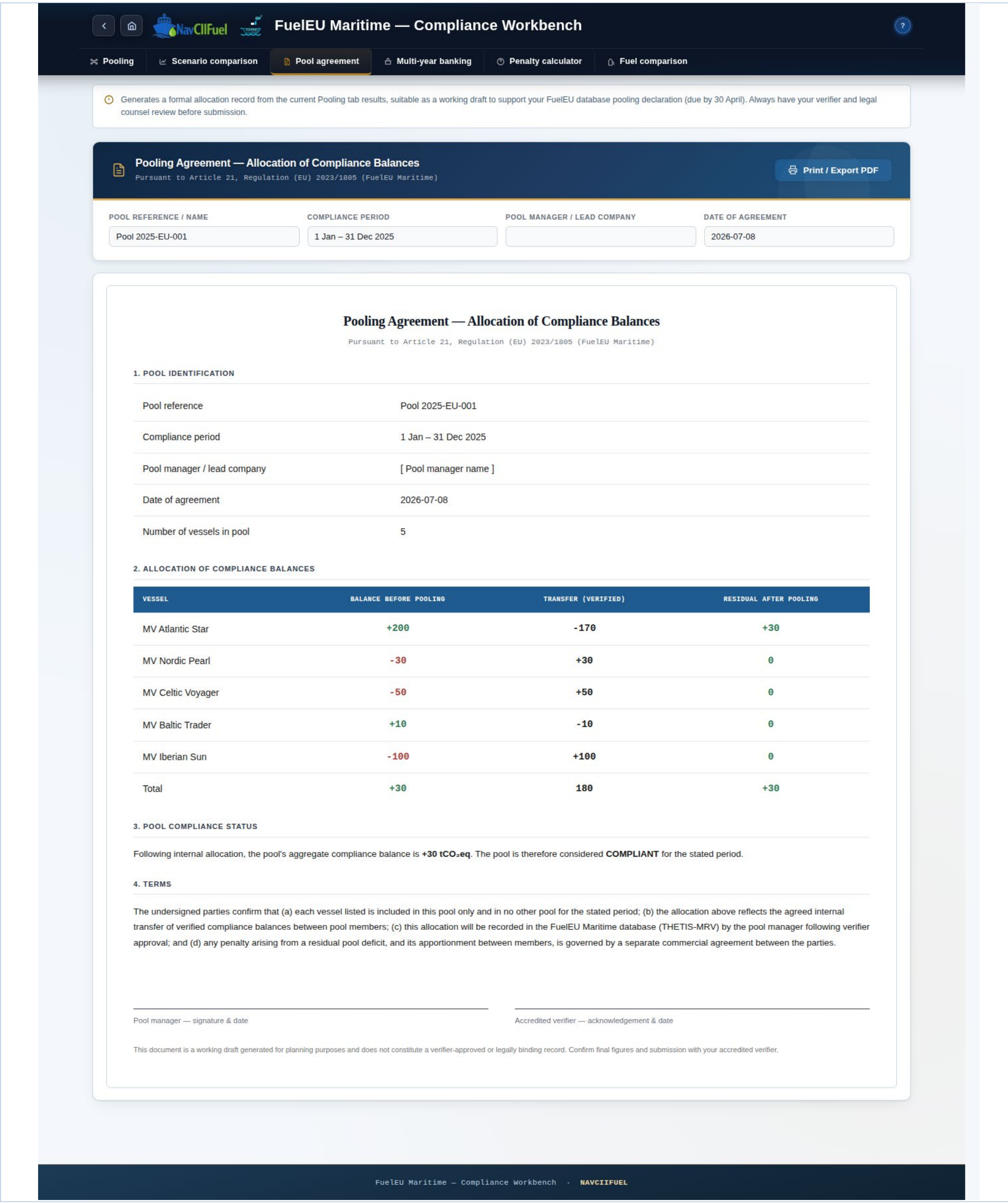


Figure 3 — Four generated allocation scenarios, ready for side-by-side comparison.

Compare the four cards side by side to choose the allocation approach that best matches your pool's commercial agreement, then use "Clear all scenarios" to start over.

6 Pool Agreement

Generates a formal draft allocation record from your current Pooling tab results — a working draft to support the FuelEU database pooling declaration (due by 30 April).



FuelEU Maritime — Compliance Workbench

Pooling · Scenario comparison · Pool agreement · Multi-year banking · Penalty calculator · Fuel comparison

Generates a formal allocation record from the current Pooling tab results, suitable as a working draft to support your FuelEU database pooling declaration (due by 30 April). Always have your verifier and legal counsel review before submission.

Pooling Agreement — Allocation of Compliance Balances

Pursuant to Article 21, Regulation (EU) 2023/1805 (FuelEU Maritime)

Print / Export PDF

POOL REFERENCE / NAME	COMPLIANCE PERIOD	POOL MANAGER / LEAD COMPANY	DATE OF AGREEMENT
Pool 2025-EU-001	1 Jan – 31 Dec 2025		2026-07-08

Pooling Agreement — Allocation of Compliance Balances

Pursuant to Article 21, Regulation (EU) 2023/1805 (FuelEU Maritime)

1. POOL IDENTIFICATION

Pool reference	Pool 2025-EU-001
Compliance period	1 Jan – 31 Dec 2025
Pool manager / lead company	[Pool manager name]
Date of agreement	2026-07-08
Number of vessels in pool	5

2. ALLOCATION OF COMPLIANCE BALANCES

VESSEL	BALANCE BEFORE POOLING	TRANSFER (VERIFIED)	RESIDUAL AFTER POOLING
MV Atlantic Star	+200	-170	+30
MV Nordic Pearl	-30	+30	0
MV Celtic Voyager	-50	+50	0
MV Baltic Trader	+10	-10	0
MV Iberian Sun	-100	+100	0
Total	+30	180	+30

3. POOL COMPLIANCE STATUS

Following internal allocation, the pool's aggregate compliance balance is **+30 tCO_2eq**. The pool is therefore considered **COMPLIANT** for the stated period.

4. TERMS

The undersigned parties confirm that (a) each vessel listed is included in this pool only and in no other pool for the stated period; (b) the allocation above reflects the agreed internal transfer of verified compliance balances between pool members; (c) this allocation will be recorded in the FuelEU Maritime database (THETIS-MRV) by the pool manager following verifier approval; and (d) any penalty arising from a residual pool deficit, and its apportionment between members, is governed by a separate commercial agreement between the parties.

Pool manager — signature & date

Accredited verifier — acknowledgement & date

This document is a working draft generated for planning purposes and does not constitute a verifier-approved or legally binding record. Confirm final figures and submission with your accredited verifier.

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Figure 4 — Draft Pooling Agreement document, generated automatically from the pool results.

Field / Control	Description
Agreement metadata fields	Pool reference/name, compliance period, pool manager/lead company, and date of agreement — fill these in to populate the generated document.
Print / Export PDF	Produces a printable version of the agreement document for circulation and signature.
Note: This is a drafting aid only — always have your verifier and legal counsel review the final agreement before submission.	

7 Multi-Year Banking

Tracks a single pool's compliance balance across multiple years, applying the regulation's banking and borrowing rules.

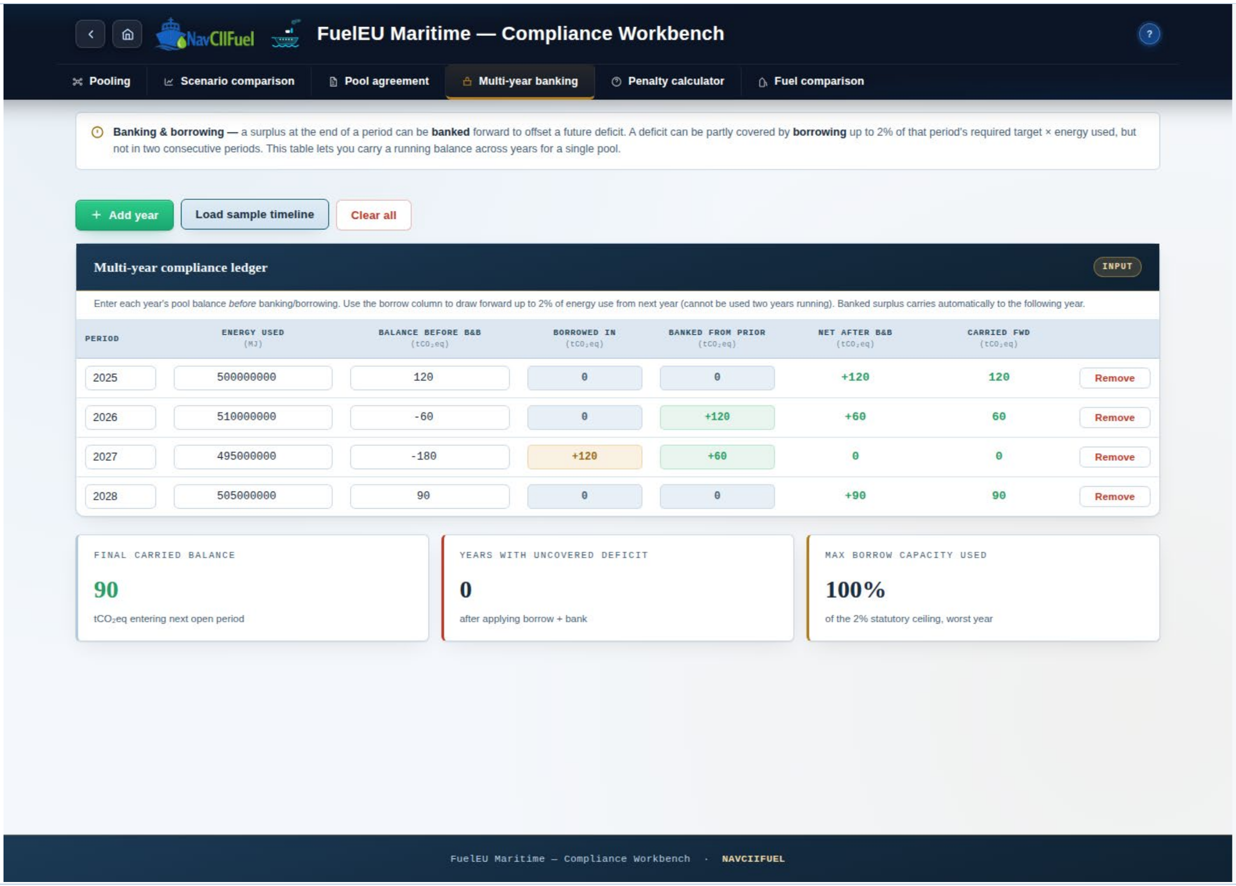


Figure 5 — Multi-year compliance ledger with a sample timeline loaded.

Field / Control	Description
Banking	A surplus at the end of a period carries forward automatically to offset a future deficit.
Borrowing	A deficit can be partly covered by borrowing up to 2% of that period's required target × energy used — but not in two consecutive periods.
Ledger table columns	Period, energy used (MJ), balance before banking/borrowing, borrowed in, banked from prior, net after banking/borrowing, and carried forward — each row is one compliance year.
Toolbar	Add year, load a sample timeline, or clear all rows.
Summary KPIs	Final carried balance entering the next open period, number of years left with an uncovered deficit, and the maximum share of the 2% statutory borrowing ceiling used in any single year.

8 Penalty Calculator

Converts a residual compliance deficit (after pooling, banking, and borrowing) into an estimated FuelEU penalty.

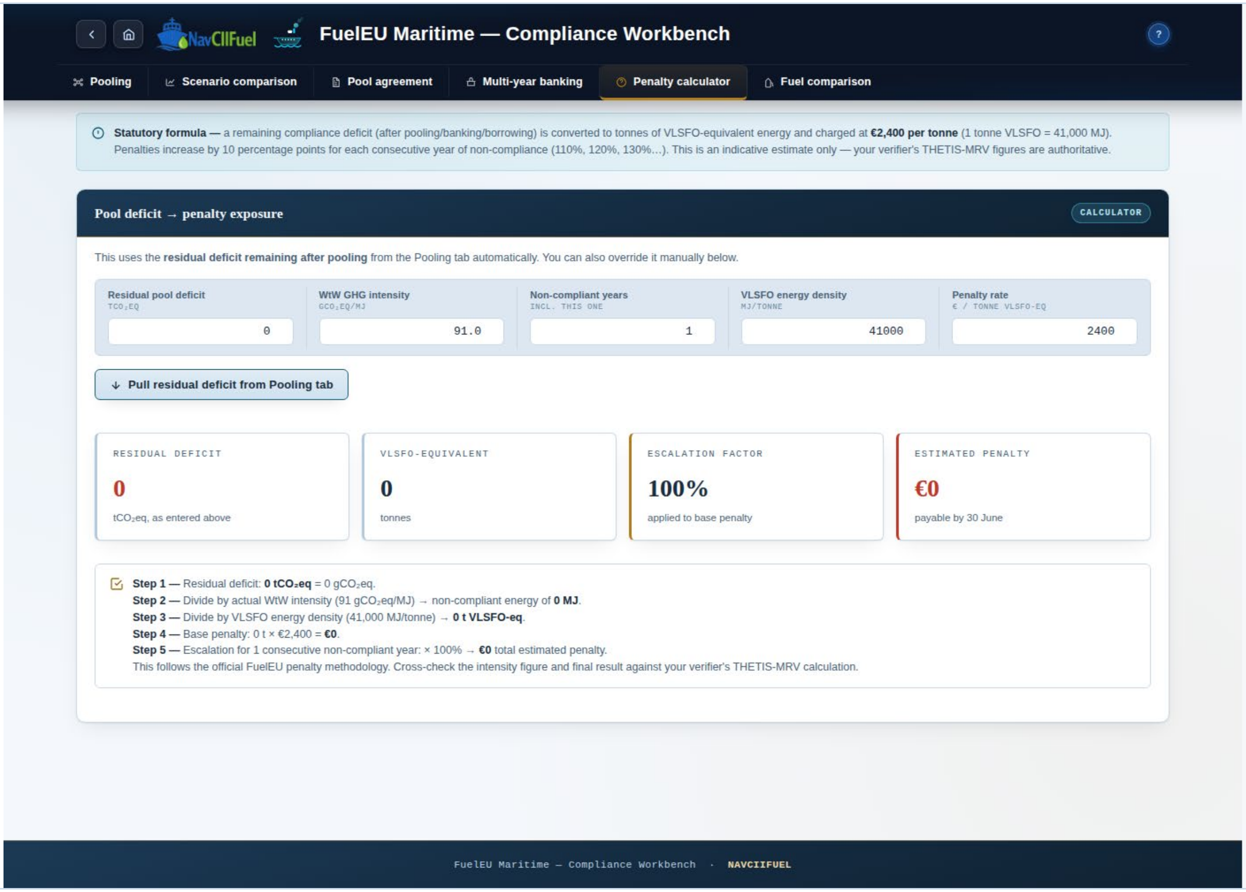


Figure 6 — Penalty calculator with the residual deficit pulled in from the Pooling tab.

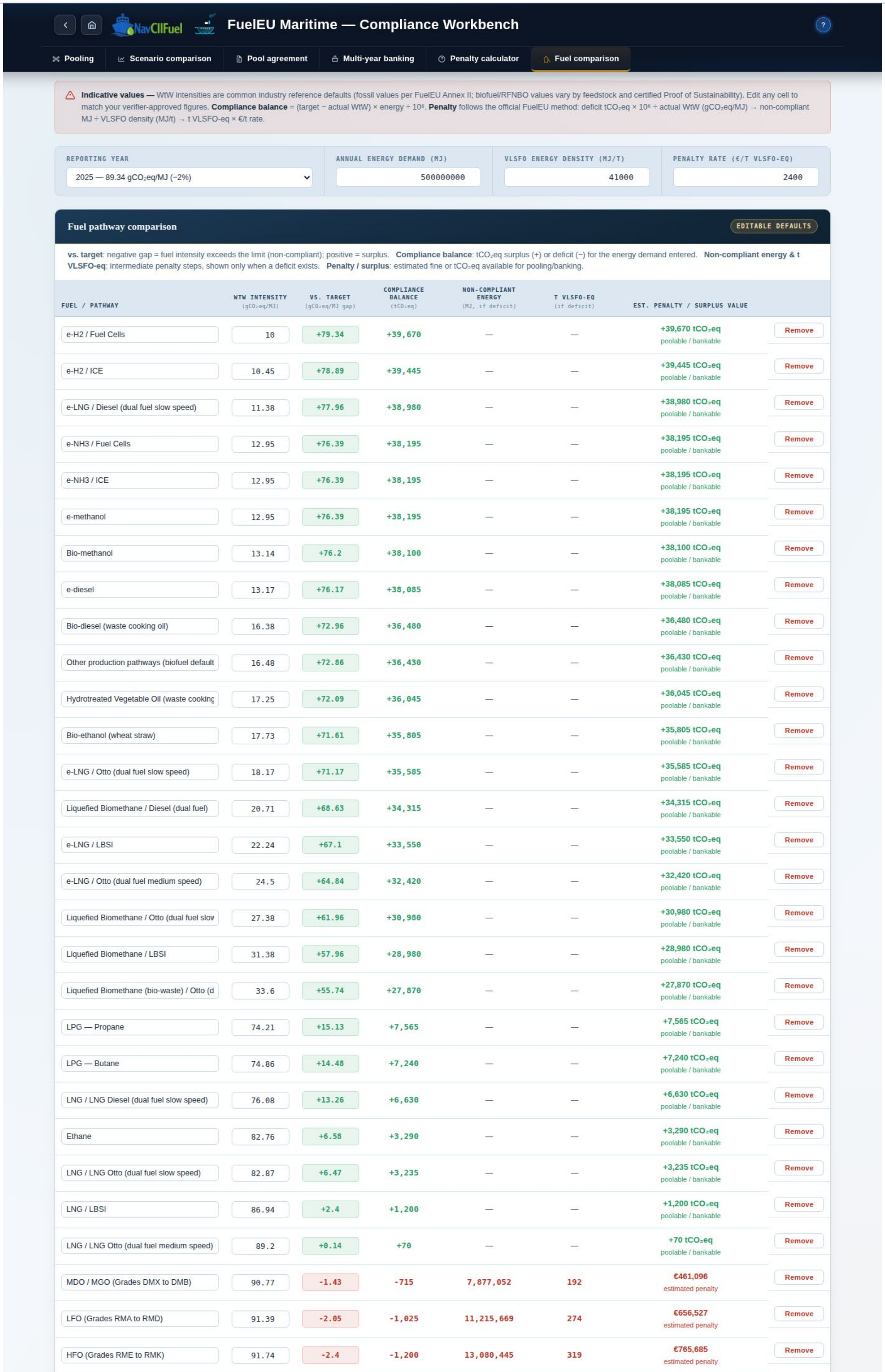
Field / Control	Description
Pull residual deficit from Pooling tab	One click imports the current uncovered deficit from the Pooling tab, instead of typing it manually.
Input fields	Residual pool deficit (tCO2eq), WtW GHG intensity (gCO2eq/MJ), number of consecutive non-compliant years, VLSFO energy density (MJ/tonne), and the penalty rate (€/tonne VLSFO-eq) — all editable.
Statutory formula	The deficit is converted to tonnes of VLSFO-equivalent energy and charged at the rate entered (default €2,400/tonne, with 1 tonne VLSFO = 41,000 MJ default). The penalty escalates by 10 percentage points for each consecutive non-compliant year (110%, 120%, 130%...).

Field / Control	Description
Result KPIs	Residual deficit, VLSFO-equivalent tonnage, the escalation factor applied, and the estimated total penalty payable — with a step-by-step breakdown beneath.

Note: This is an indicative estimate only — your verifier's THETIS-MRV figures are authoritative for actual filing.

9 Tab 6 — Fuel Comparison

Compares multiple fuel pathways' well-to-wake (WtW) GHG intensity against the FuelEU compliance target for a chosen reporting year.



FUEL / PATHWAY	WTW INTENSITY (gCO₂eq/MJ)	VS. TARGET (gCO₂eq/MJ gap)	COMPLIANCE BALANCE (tCO₂eq)	NON-COMPLIANT ENERGY (MJ, if deficit)	T VLSFO-EQ (if deficit)	EST. PENALTY / SURPLUS VALUE	
e-H2 / Fuel Cells	10	+79.34	+39,670	—	—	+39,670 tCO₂eq poolable / bankable	Remove
e-H2 / ICE	10.45	+78.89	+39,445	—	—	+39,445 tCO₂eq poolable / bankable	Remove
e-LNG / Diesel (dual fuel slow speed)	11.38	+77.96	+38,980	—	—	+38,980 tCO₂eq poolable / bankable	Remove
e-NH3 / Fuel Cells	12.95	+76.39	+38,195	—	—	+38,195 tCO₂eq poolable / bankable	Remove
e-NH3 / ICE	12.95	+76.39	+38,195	—	—	+38,195 tCO₂eq poolable / bankable	Remove
e-methanol	12.95	+76.39	+38,195	—	—	+38,195 tCO₂eq poolable / bankable	Remove
Bio-methanol	13.14	+76.2	+38,100	—	—	+38,100 tCO₂eq poolable / bankable	Remove
e-diesel	13.17	+76.17	+38,085	—	—	+38,085 tCO₂eq poolable / bankable	Remove
Bio-diesel (waste cooking oil)	16.38	+72.96	+36,480	—	—	+36,480 tCO₂eq poolable / bankable	Remove
Other production pathways (biofuel default	16.48	+72.86	+36,430	—	—	+36,430 tCO₂eq poolable / bankable	Remove
Hydrotreated Vegetable Oil (waste cooking	17.25	+72.09	+36,045	—	—	+36,045 tCO₂eq poolable / bankable	Remove
Bio-ethanol (wheat straw)	17.73	+71.61	+35,805	—	—	+35,805 tCO₂eq poolable / bankable	Remove
e-LNG / Otto (dual fuel slow speed)	18.17	+71.17	+35,585	—	—	+35,585 tCO₂eq poolable / bankable	Remove
Liquefied Biomethane / Diesel (dual fuel)	20.71	+68.63	+34,315	—	—	+34,315 tCO₂eq poolable / bankable	Remove
e-LNG / LBSI	22.24	+67.1	+33,550	—	—	+33,550 tCO₂eq poolable / bankable	Remove
e-LNG / Otto (dual fuel medium speed)	24.5	+64.84	+32,420	—	—	+32,420 tCO₂eq poolable / bankable	Remove
Liquefied Biomethane / Otto (dual fuel slov	27.38	+61.96	+30,980	—	—	+30,980 tCO₂eq poolable / bankable	Remove
Liquefied Biomethane / LBSI	31.38	+57.96	+28,980	—	—	+28,980 tCO₂eq poolable / bankable	Remove
Liquefied Biomethane (bio-waste) / Otto (d	33.6	+55.74	+27,870	—	—	+27,870 tCO₂eq poolable / bankable	Remove
LPG — Propane	74.21	+15.13	+7,565	—	—	+7,565 tCO₂eq poolable / bankable	Remove
LPG — Butane	74.86	+14.48	+7,240	—	—	+7,240 tCO₂eq poolable / bankable	Remove
LNG / LNG Diesel (dual fuel slow speed)	76.08	+13.26	+6,630	—	—	+6,630 tCO₂eq poolable / bankable	Remove
Ethane	82.76	+6.58	+3,290	—	—	+3,290 tCO₂eq poolable / bankable	Remove
LNG / LNG Otto (dual fuel slow speed)	82.87	+6.47	+3,235	—	—	+3,235 tCO₂eq poolable / bankable	Remove
LNG / LBSI	86.94	+2.4	+1,200	—	—	+1,200 tCO₂eq poolable / bankable	Remove
LNG / LNG Otto (dual fuel medium speed)	89.2	+0.14	+70	—	—	+70 tCO₂eq poolable / bankable	Remove
MDO / MGO (Grades DMX to DMB)	90.77	-1.43	-715	7,877,052	192	€461,096 estimated penalty	Remove
LFO (Grades RMA to RMD)	91.39	-2.05	-1,025	11,215,669	274	€656,527 estimated penalty	Remove
HFO (Grades RME to RMK)	91.74	-2.4	-1,200	13,080,445	319	€765,685 estimated penalty	Remove

Figure 7 — Fuel pathway comparison table and GHG intensity chart against the reporting-year target.

Field / Control	Description
Parameter bar	Reporting year (2025-2050, each with its own target intensity and required reduction %), annual energy demand (MJ), VLSFO energy density (MJ/t), and the penalty rate (€/t VLSFO-eq).
Fuel pathway table	For each fuel/pathway: WtW intensity, gap vs. target, compliance balance (tCO2eq), non-compliant energy and VLSFO-equivalent tonnes (shown only where there is a deficit), and the estimated penalty or surplus value. Add a custom fuel row or reset the table to its defaults at any time.
GHG intensity chart	A bar chart of each fuel's WtW intensity against a dashed line marking the year's target — bars to the left of the line are compliant (surplus), bars to the right are non-compliant (deficit).

Note: WtW intensities shown by default are common industry reference values (fossil per FuelEU Annex II; biofuel/RFNBO values vary by feedstock and certified Proof of Sustainability) — edit any cell to match your verifier-approved figures.

10 Getting Help In-App

Click the ? button in the masthead at any time to open the built-in help panel. It contains a quick-link table of contents and a description of every field across all six tabs, plus an Important Notes section on the tool's scope and limits.

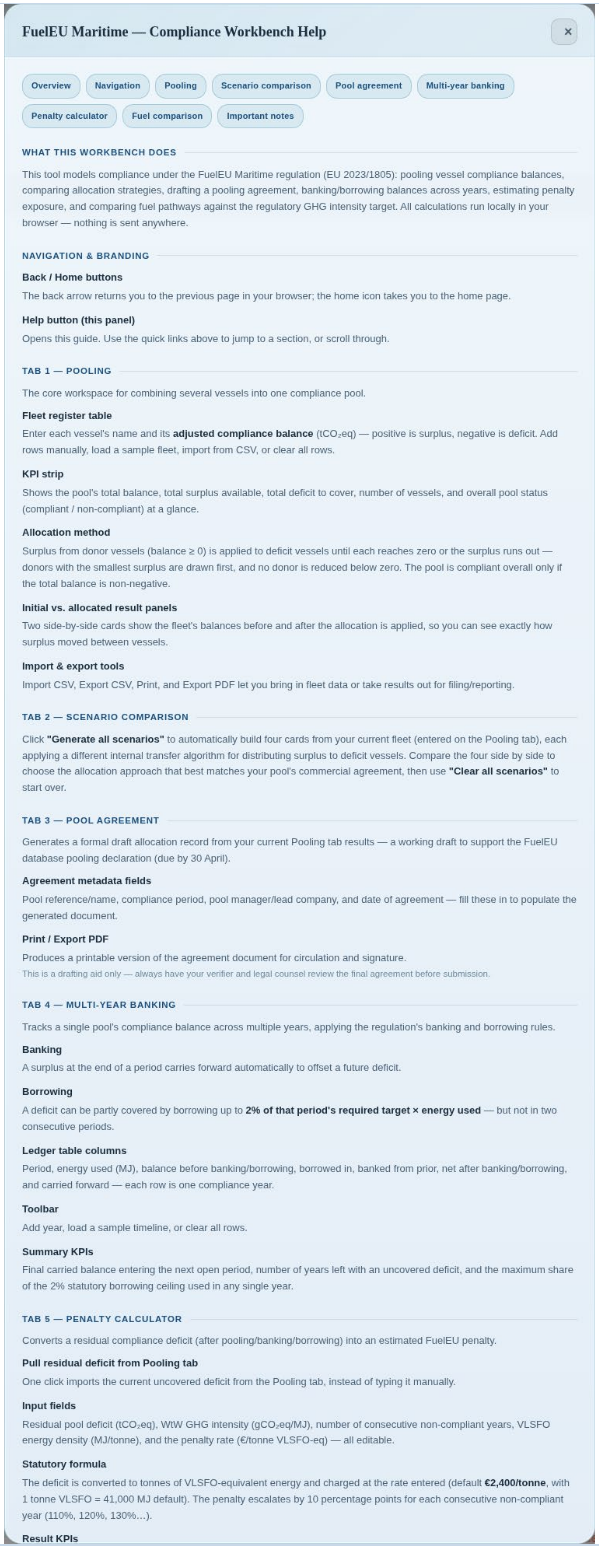


Figure 8 — In-app help panel, opened from the ? button in the masthead.

11 Best-Practice Checklist

- Keep vessel names and reporting periods consistent across the Pooling, Banking, and Agreement tabs.
- Confirm each vessel's adjusted compliance balance is verifier-approved before pooling.
- Check WtW intensity and CF values before relying on the Fuel comparison table, especially for custom or biofuel entries.
- Review the pool status and residual deficit before generating the Pool Agreement or estimating penalties.
- Re-run "Generate all scenarios" after any change to the fleet register, since scenarios do not update automatically.
- Store exported CSV/PDF files together with the generated Pool Agreement for full traceability.

12 Troubleshooting

Symptom	Likely cause / fix
Scenario cards do not appear	"Generate all scenarios" has not been clicked yet, or the fleet register on the Pooling tab is empty — add vessels first, then generate scenarios.
Pool Agreement document looks outdated	The agreement regenerates automatically each time you open the Pool agreement tab or edit a metadata field — switch tabs away and back, or edit any metadata field, to refresh it.
Penalty calculator shows 0	Click "Pull residual deficit from Pooling tab", or check that the Pooling tab actually has an uncovered deficit (a fully compliant pool has no penalty).
Uploaded CSV did not populate all rows	Column headers in the file do not match the expected format — check against the Export CSV template and correct column names.
Fuel chart looks outdated	Change the Reporting Year or edit a fuel row to refresh the chart; it recalculates immediately on any parameter change.
Borrowing not applied in the banking ledger	Borrowing cannot be used in two consecutive periods, and is capped at 2% of that period's required target × energy used — check the prior year's row.
Print / Export PDF opens a blank or partial page	Allow the browser a few seconds to render before printing, and ensure pop-ups are not blocked for this file.

End of user guide.